Joshua Brown
Web & Cloud
Labs7

7.1a: Terraform AWS Guestbook

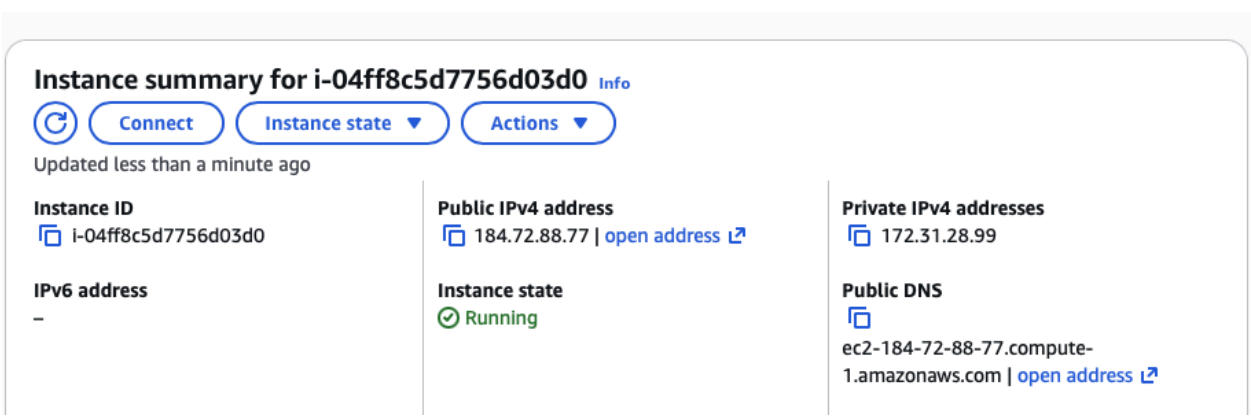
1-4

Take a screenshot showing the completion of the command including its output

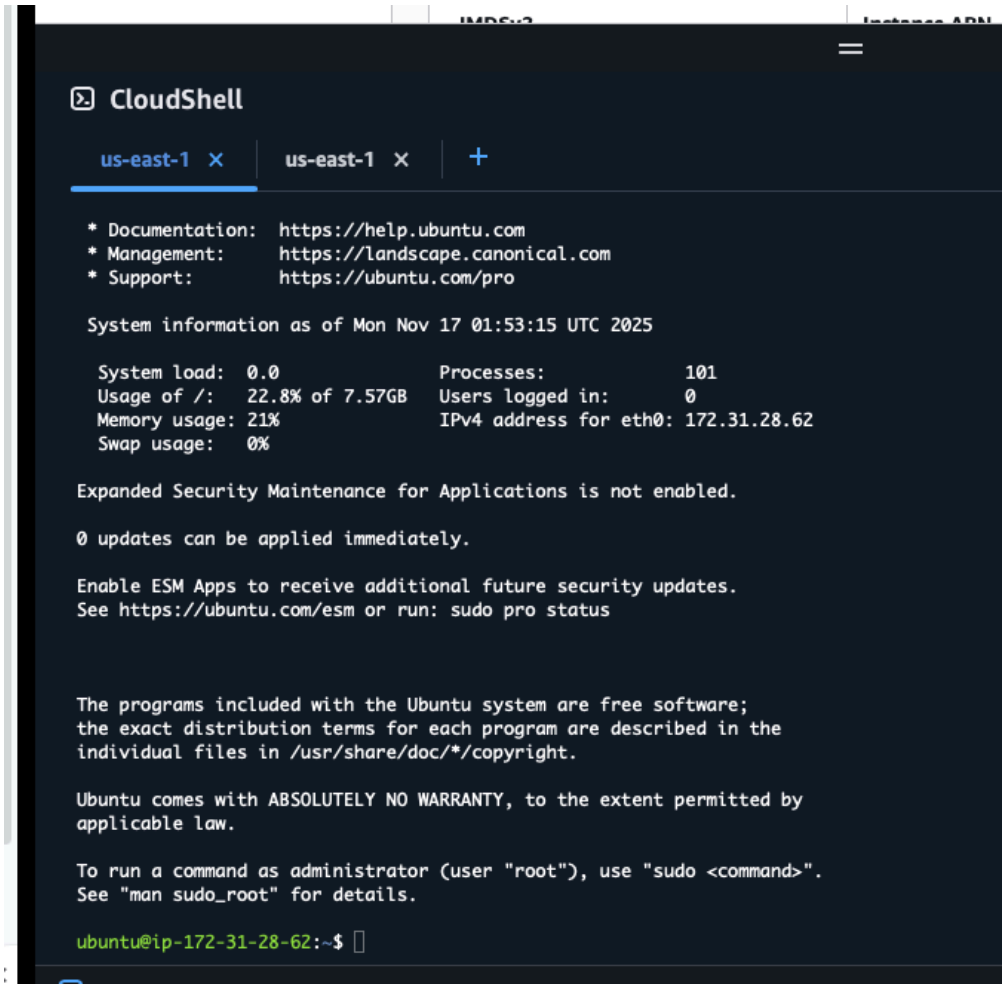
```
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + ec2instance = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes
                                        AWS CloudShell
aws_instance.guestbook: Creating...
aws_instance.guestbook: Still creating... [00m10s elapsed]
aws_instance.guestbook: Creation complete after 13s [id=i-04ff8c5d7756d03d0]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

ec2instance = "184.72.88.77"
tf $
```

Visit EC2 within the web console and refresh it to see that the IP address has been bound to the VM.

Take a screenshot that includes the VM's IP addresses



5-6Take a screenshot of the successful ssh login from Cloud Shell.



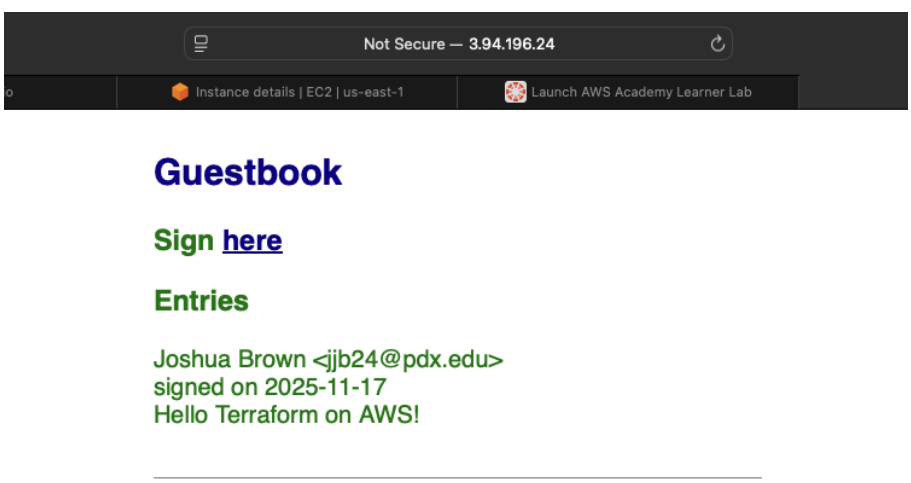
7

Take a screenshot of the output of the command that includes the IP address of the instance

```
Changes to Outputs:
  + ec2instance = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_key_pair.kp: Creating...
aws_security_group.sg-guestbook: Creating...
aws_key_pair.kp: Creation complete after 0s [id=guestbook-key]
aws_security_group.sg-guestbook: Creation complete after 3s [id=sg-031241f5804cdbd35]
aws_instance.guestbook: Creating...
aws_instance.guestbook: Still creating... [00m10s elapsed]
aws_instance.guestbook: Creation complete after 12s [id=i-0e3c9f2f47da3ea2c]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

ec2instance = "3.94.196.24"
tf $
```

3.94.196.24

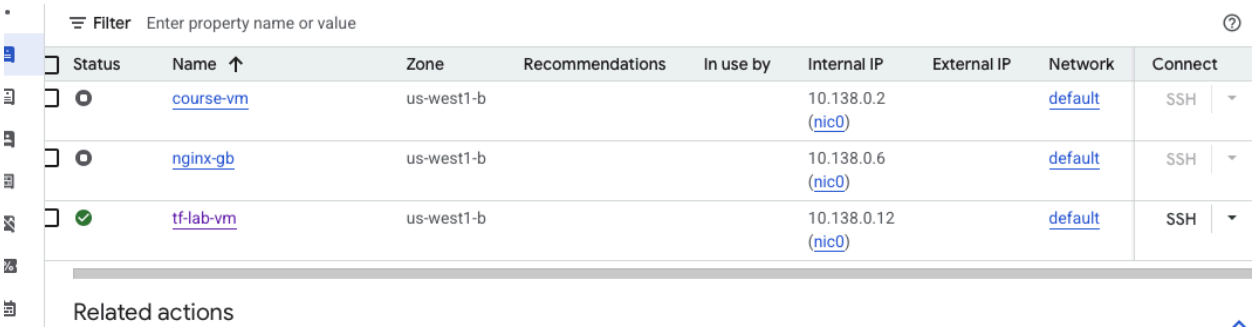
8. Take a screenshot of the Guestbook including the URL with the entry in it.



7.1g - Terraform GCP

1-4

Take a screenshot that includes the VM's IP addresses



Filter Enter property name or value

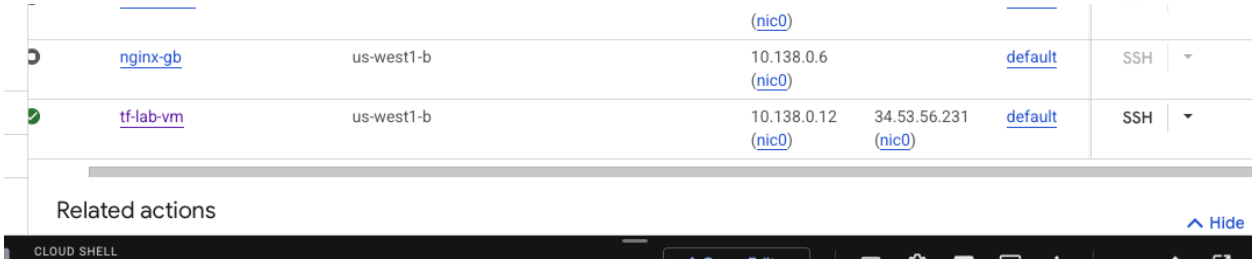
Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Network	Connect
	course-vm	us-west1-b			10.138.0.2 (nic0)		default	SSH
	nginx-gb	us-west1-b			10.138.0.6 (nic0)		default	SSH
	tf-lab-vm	us-west1-b			10.138.0.12 (nic0)		default	SSH

Related actions

5 - Take a screenshot showing the completion of the command including its output

```
google_compute_address.static: Creation complete after 11s [id=projects/cloud-brown-
pv4-address]
google_compute_instance.default: Modifying... [id=projects/cloud-brown-jjb24/zones/
google_compute_instance.default: Still modifying... [id=projects/cloud-brown-jjb24/
m, 10s elapsed]
google_compute_instance.default: Modifications complete after 11s [id=projects/clou
ances/tf-lab-vm]

Apply complete! Resources: 1 added, 1 changed, 0 destroyed.

Outputs:

ip = "34.53.56.231"
jjb24@cloudshell:~/tf (cloud-brown-jjb24)$
```

Take a screenshot that includes the VM's IP addresses



			(nic0)			
nginx-gb	us-west1-b	10.138.0.6 (nic0)		default	SSH	
tf-lab-vm	us-west1-b	10.138.0.12 (nic0)	34.53.56.231 (nic0)	default	SSH	

Related actions

Hide

CLOUD SHELL

6

- Take a screenshot of the successful ssh login from Cloud Shell.

```
jjb24@cloudshell:~/tf (cloud-brown-jjb24)$ ssh 34.53.56.231
The authenticity of host '34.53.56.231 (34.53.56.231)' can't be established
ED25519 key fingerprint is SHA256:LXH0JjQBrbioARJtlH65hM3aYRThK01FE1bx6vQTN
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.53.56.231' (ED25519) to the list of known ho
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1043-gcp x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Nov 18 04:25:02 UTC 2025

  System load:  0.0               Processes:             104
  Usage of /:   22.7% of 9.51GB   Users logged in:       0
  Memory usage: 6%                IPv4 address for ens4: 10.138.0.12
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

jjb24@tf-lab-vm:~$
```

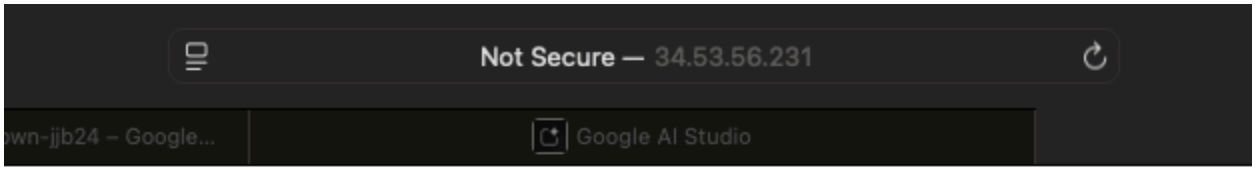
7

What resources are being added, changed, or destroyed?
The resource replaced is google_compute_instance.default

What part of the configuration forces a replacement to occur?

```
+ metadata_startup_script = <<-EOT # forces replacement
      #!/bin/bash
      apt update
      apt install -yq build-essential python3-pip python3-flask gunicorn git
      git clone https://github.com/wu4f/cs430-src /root/cs430-src
      cd /root/cs430-src/03_nginx_gunicorn_certbot
      gunicorn --bind :80 --workers 1 --threads 8 app:app
  EOT
+ min_cpu_platform        = (known after apply)
```

8 -Take a screenshot of the Guestbook including the URL with the entry in it.



Guestbook

Sign here

Entries

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-18
Hello Terraform on GCP!

7.2g

1-4
What is the name of the Instance Template dynamically generated to create the two nodes (VMs)?
gke-guestbook-default-pool-fa8e0e33

What is the name of the Instance Group dynamically generated that the two nodes belong to?
gke-guestbook-default-pool-fa8e0e33-grp

What are the names of the two nodes?
gke-guestbook-default-pool-fa8e0e33-0s91
gke-guestbook-default-pool-fa8e0e33-cffk

5

Take a screenshot of the container image created

Artifact Registry / ··· / Location: us / Repository: gcr.io / Package: gcp_gb / Version: sha256:7b8cd6c30986982ee380a6

7b8cd6c309869 | Delete | Setup instructions | Deploy | Refresh | Copy pat

Overview | Dependencies | Vulnerabilities | Pull | Manifest | Files | Attachments

Format	Docker
Media type	application/vnd.docker.distribution.manifest.v2+json
Project	cloud-brown-jjb24
Location	us (multiple regions in United States)
Repository	gcr.io
Image	gcp_gb
Digest	sha256:7b8cd6c30986982ee380a6cc385b366c5df32a056b573629ce9af3a29c10f117
Virtual size	1.4 GB
Built	Nov 17, 2025, 8:55:15 PM
Created	Nov 17, 2025, 9:24:12 PM
Updated	Nov 17, 2025, 9:24:12 PM
Tags	latest
Build	API disabled
Subject digest	—
Artifact type	—

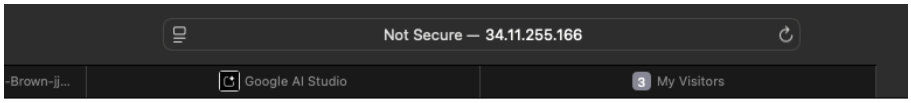
Take a screenshot of the output of the following command when all 3 replicas reach a "Running" state.

```
jjb24@cloudshell:~/cs430-src/05_gcp_datastore (cloud-brown-jjb24)$ kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
guestbook-replicas-jz5lq    1/1     Running   0          2m9s
guestbook-replicas-vgtb5    1/1     Running   0          2m9s
guestbook-replicas-wszz4    1/1     Running   0          2m9s
jjb24@cloudshell:~/cs430-src/05_gcp_datastore (cloud-brown-jjb24)$
```

Take a screenshot of listing services with LoadBalancer indicating an external IP address that is ready for access.

```
jjb24@cloudshell:~/cs430-src/05_gcp_datastore (cloud-brown-jjb24)$ kubectl get services
NAME           TYPE           CLUSTER-IP       EXTERNAL-IP     PORT(S)        AGE
guestbook-lb   LoadBalancer   34.118.225.121   34.11.255.166   80:31661/TCP   2m52s
kubernetes     ClusterIP      34.118.224.1     <none>          443/TCP        43m
jjb24@cloudshell:~/cs430-src/05_gcp_datastore (cloud-brown-jjb24)$
```

Take a screenshot of the Guestbook including the URL with the entry in it.



Guestbook

Sign here

Entries

Joshua Brown <jjb24@pdx.e3du>
signed on 2025-11-04 07:10:59.615802+00:00
Hello Cloud Shell!

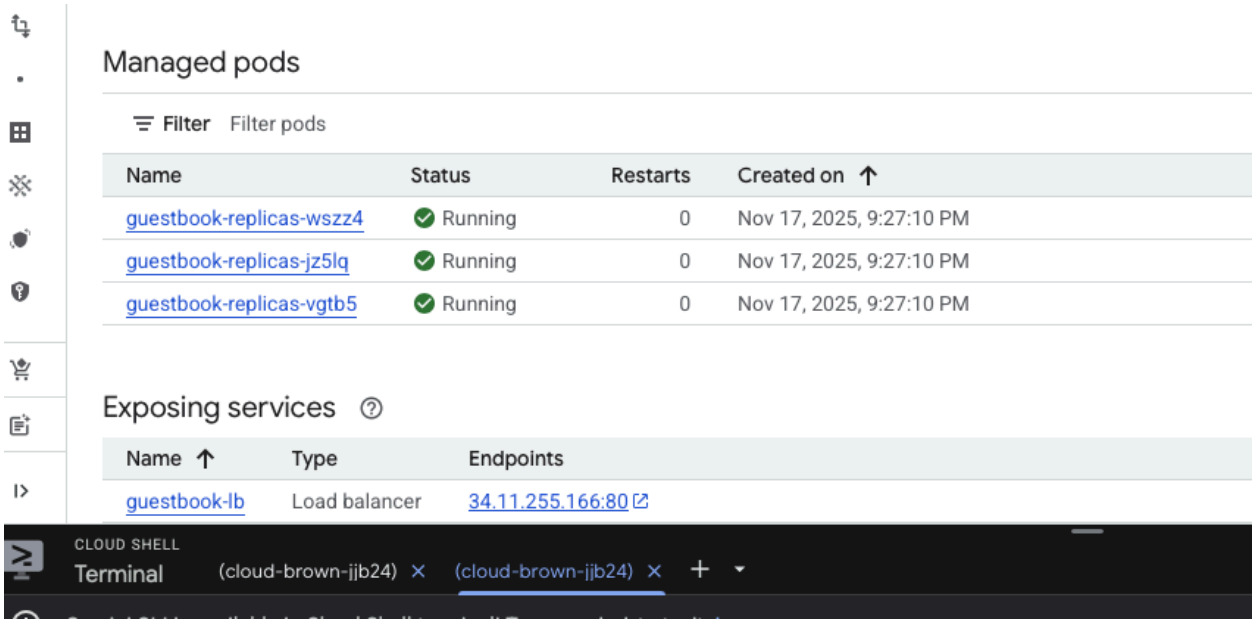
Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-04 06:42:10.129652+00:00
Hello Datastore

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-05 20:07:11.711272+00:00
Hello App Engine!

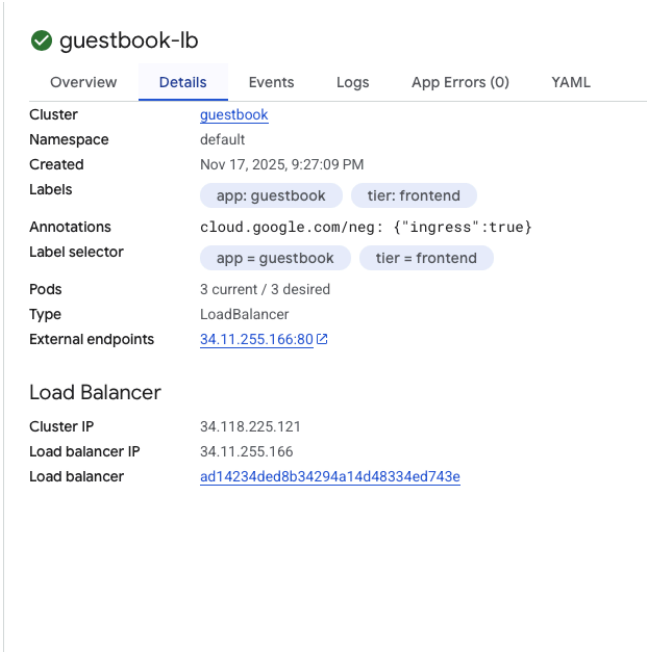
Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-04 07:18:41.535992+00:00
Hello Compute Engine!

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-18 05:31:07.186291+00:00
Hello Kubernetes!

- Take a screenshot of the managed guestbook pods and the service being exposed.



Take a screenshot of the load balancer and its details



- Take a screenshot of the addresses allocated and indicate the ones associated with nodes versus the one associated with the load balancer.
    - load balancer:

Load Balancer

Cluster IP	34.118.225.121
Load balancer IP	34.11.255.166

- nodes

34.11.255.166:80

Take a screenshot of the Guestbook including the URL with the entry in it.

Not Secure — 34.11.255.166

Google AI Studio

Guestbook

Sign here

Entries

Joshua Brown <jjb24@pdx.e3du>
signed on 2025-11-04 07:10:59.615802+00:00
Hello Cloud Shell!

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-04 06:42:10.129652+00:00
Hello Datastore

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-05 20:07:11.711272+00:00
Hello App Engine!

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-04 07:18:41.535992+00:00
Hello Compute Engine!

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-18 05:31:07.186291+00:00
Hello Kubernetes!

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-04 06:57:18.891004+00:00
Hello Docker Datastore!

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-10 21:33:09.912997+00:00
Hello Cloud Run!

Joshua Brown <jjb24@pdx.edu>
signed on 2025-11-18 05:55:25.673060+00:00
Hello Cloud Build!

7.3g

- Does Google provide a Python package specifically for accessing the Knowledge Graph API?

No, Google does not provide a Python package specifically for the Knowledge Graph Search API.

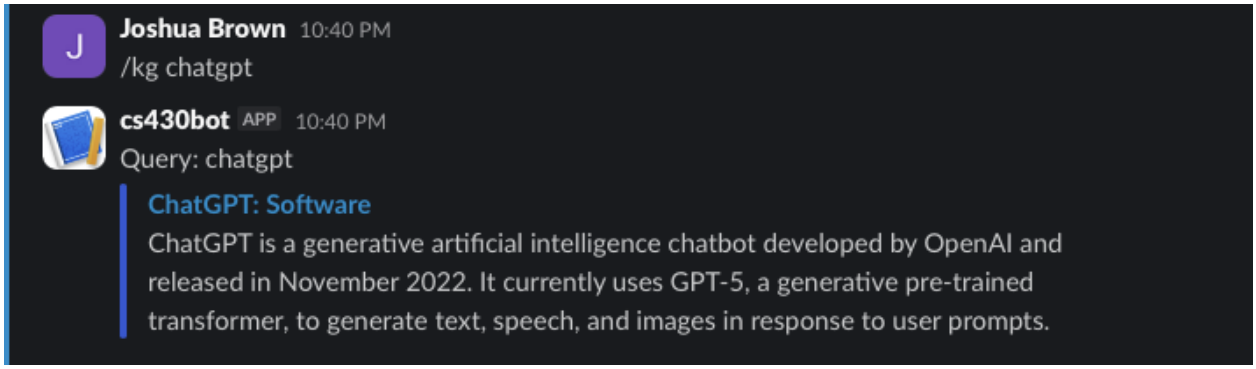
- Show the source line that constructs the query we wish to send to the Knowledge Graph API.
    - req = kgsearch.entities().search(query=query, limit=1)
- Show the source line that then executes the query and saves the response. What is the name of the method that sends the query to the Knowledge Graph API?
    - res = req.execute()
        - the name of the method that sends the query is .execute()
- What is the Python data type that is used to represent the formatted message?
    - dict
- What are the three main attributes of the formatted message passed back to Slack?
    - response_type
    - text
    - attachments

kg api key: AIzaSyA6bN0zLujtny2NE-PMoZR4zVJUDmbn0O8

slack secret: 7f061d1f86c8a12cb0159d30c278954d

note the url: https://us-central1-cloud-brown-jjb24.cloudfunctions.net/kg_search

Take a screenshot of its response for your lab notebook



7.4g - Show the output for your lab notebook

```
ice account info is missing 'email' field.
(env) jjb24@cloudshell:~/python-docs-samples/vision/snippets/detect (cloud-brown-jjb24)$ python detect.py lab
pi-codelab/birds.jpg
Labels:
Bird
Ratite
Common ostrich
Flightless bird
Beak
Vertebrate
Wildlife
Terrestrial animal
Greater rhea
Feather
(env) jjb24@cloudshell:~/python-docs-samples/vision/snippets/detect (cloud-brown-jjb24)$
Display a menu
```

- What is the name of the function?
  - detect_labels_uri
- What type of Vision client is instantiated in it?
  - vision.imageannotatorclient
- What method is invoked in the Vision client to perform the detection?
  - label_detection
- What is the name of the attribute in the response object that contains the results we seek?
  - label_annotations
- Take a screenshot of the output for the above commands
  -

```
(env) jjb24@cloudshell:~/python-docs-samples/vision/snippets/detect (cloud-brown-jjb24)$ python detect.py logos psulogo.jpg
Logos:
Portland State University
(env) jjb24@cloudshell:~/python-docs-samples/vision/snippets/detect (cloud-brown-jjb24)$
Display a menu
```

- What method is invoked in the Vision client to perform the detection?
  - logo_detection()

4 - Show the output for your lab notebook

```
(env) jjb24@cloudshell:~/python-docs-samples/speech/snippets (cloud-br
Transcript: how old is the Brooklyn Bridge
(env) jjb24@cloudshell:~/python-docs-samples/speech/snippets (cloud-br
```

- What is the name of the function?
  - transcribe_file
- What method is invoked in the Speech client to perform the detection?
  - recognize()
- What is the name of the attribute in the response object that contains the results we seek?
  - results

- Show the output for your lab notebook
-

```
[notice] To update, run: pip install --upgrade pip
(env) jjb24@cloudshell:~/python-docs-samples/speech/snippets (cloud-brown-jjb24)$ cd ~/python-docs-samples/translate/samples/s
(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$ python snippets.py translate-text
'
Detected source language: zh-TW
Input text: 你有沒有帶外套
Translated text: Did you bring a jacket?

(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$
```

open up snippets.py

- What is the name of the function?
  - translate-text
- What method is invoked in the Translate client to perform the detection?
  - translate
- What is the name of the attribute in the response object that contains the results we seek?
  - translatedText

6 - Show the output for your lab notebook

```
(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$
"homework is awful!" has sentiment=-0.800000011920929

Entities are:
name: homework
"homework is ok" has sentiment=0.30000001192092896

Entities are:
name: homework
"homework is awesome?" has sentiment=0.4000000059604645

Entities are:
name: homework
"homework is awesome!" has sentiment=0.8999999761581421

Entities are:
name: homework
"The protestors in Oregon put on gas masks and wore yellow t-shirts" has sentiment=-0.600000023841

Entities are:
name: protestors
name: gas masks
name: Oregon
name: t-shirts
(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$
```

7 -

What is the name of the function that performs the transcription?
transcribe_gcs

- What is the name of the function that performs the translation?
  - translate_text
- What is the name of the function that performs the entity analysis on the translation?
  - entities_text
- What is the name of the function that performs the entity analysis on the image?
  - detext_labels_uri

9 - If the program deems them unrelated, then based on the results from the APIs, what must be changed in the program to address this?

```
(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$
Transcription: willst du mit uns Fußball spielen
Translation: Do you want to play football with us?
Entities: ['football']
Image labels: ['Ball', 'Football', 'Sports', 'Football', 'Ball game', 'Soccer ball', 'Sports venue', 'Soccer-specific stadium']
The audio and image do not appear to be related.
(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$
Display a menu
```

the program must be changed to perform a case-insensitive comparison

- If the program deems them unrelated, then based on the results from the APIs, what must be changed in the program to address this?
  -
```
(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$
Transcription: bisikletimi sokağa bırak
Translation: leave my bike on the street
Entities: ['bike', 'street']
Image labels: ['Bicycle', 'Bicycle frame', 'Bicycle handlebar', 'Bicycle tire', 'Bicycle Wheel Rim', 'Bicycle chain', 'Bicycle wheel', 'Spoke', '
Bicycle fork', 'Crankset']
The audio and image do not appear to be related.
(env) jjb24@cloudshell:~/python-docs-samples/translate/samples/snippets (cloud-brown-jjb24)$
Display a menu
```
  - the program's comparison logic must be improved to handle synonyms or related concepts
- If the program deems them unrelated, then based on the results from the APIs, what must be changed in the program to address this?
  - the program's comparison logic needs to be made more flexible to handle pluralization and partial matches

13 - What are the 3 labels with the highest confidence that the Video Intelligence API associates with the video and what are the confidences for each?

1. **sports:** Confidence: 0.9218811392784119
2. **basketball:** Confidence: 0.9137870669364929
3. **player:** Confidence: 0.8446521162986755

- What is the name of the client class in the package that is used?
  - videointelligenceserviceclient
- What method is used in that class to perform the annotation?
  - annotate_video

Take a screenshot for your lab notebook that includes the url

- What line of code creates the query for previous detections?
  - query = datastore_client.query(kind="Faces")
- What line of code sends the query to Cloud Datastore?
  - image_entities = list(query.fetch())
- Show the line that retrieves the name of the storage bucket to use.
  - CLOUD_STORAGE_BUCKET = os.environ.get("CLOUD_STORAGE_BUCKET")
- What form field is used to specify the uploaded photo?
  - file
- Show the line that copies the photo's contents to the storage bucket.
  - blob.upload_from_string(photo.read(), content_type=photo.content_type)
- What method in Vision's annotation client is used to perform the analysis?
  - face_detection
- What fields are stored in Cloud Datastore for each image?
  - `blob_name`, `image_public_url`, `timestamp`, and `joy`.
- What happens at the end of the upload_photo route?
  - I got redirected back to the homepage